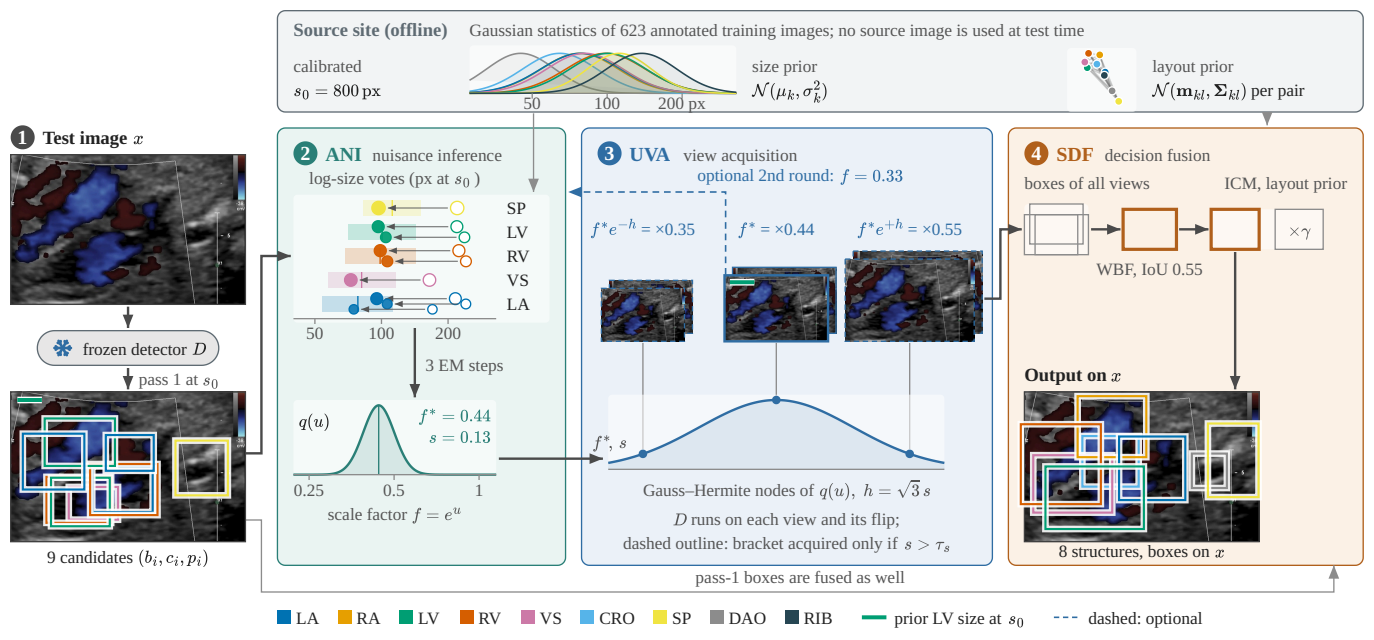

Source site (offline)
Gaussian statistics of 623 annotated training images; no source image is used at test time
calibrated
$s_0 = 800$ px
50
100
200 px
size prior
$\mathcal{N}(\mu_k, \sigma_k^2)$
layout prior
$\mathcal{N}(\mathbf{m}_{kl}, \mathbf{\Sigma}_{kl})$ per pair
1 Test image x
frozen detector D
pass 1 at s_0
9 candidates (b_i, c_i, p_i)
2 ANI nuisance inference
log-size votes (px at s_0)
SP
LV
RV
VS
LA
50
100
200
3 EM steps
$q(u)$
$f^* = 0.44$
$s = 0.13$
0.25
0.5
1
scale factor $f = e^u$
3 UVA view acquisition
optional 2nd round: $f = 0.33$
$f^* e^{-h}$ = ×0.35
f^* = ×0.44
$f^* e^{+h}$ = ×0.55
f^*, s
Gauss–Hermite nodes of $q(u)$, $h = \sqrt{3}\, s$
D runs on each view and its flip;
dashed outline: bracket acquired only if $s > \tau_s$
4 SDF decision fusion
boxes of all views
ICM, layout prior
×γ
WBF, IoU 0.55
Output on x
8 structures, boxes on x
pass-1 boxes are fused as well
LA
RA
LV
RV
VS
CRO
SP
DAO
RIB
prior LV size at s_0
dashed: optional